YCCify - Yeshwantrao Chavan College of Engineering

IT701 - Machine Learning | Dr. S. S. Khedkar

Unit 3 - Classification Algorithms

Topic: Decision Trees, SVM & Naive Bayes

3.1 Introduction

Classification is a supervised learning technique which is used to predict the category of a given input.

3.2 Common Algorithms:

* Decision Tree Entropy, Information Gain, Gini Impurity
* Naive Bayes Bayes Theorem with conditional independence
* Support Vector Machine SVM, Margin Maximization, Kernel Trick
* k-Nearest Neighbour k-NN, Euclidean & Manhattan metrics

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3.3 Model Evaluation Metrics

Evaluating classification performance requires multiple complementary metrics.

Metrics:

* Confusion Matrix TP, FP, TN, FN
* Accuracy, Precision, Recall, F1-Score
* ROC Curve and Area Under Curve AUC